



VP AIR TEST NEET (2025-26)

PART TEST - 05

DROPPER

DURATION: 180 Minutes

Date: 16-11-2025

M. MARKS: 720

Topics Covered

Physics:	Mechanical Properties of Fluids, Thermal Properties of Matter, Kinetic Theory, Thermodynamics
Chemistry:	Ionic Equilibrium, Electrochemistry, Chemical Kinetics, Organic Chemistry- IUPAC Nomenclature
Botany:	Biological Classification, Microbes in Human Welfare
Zoology:	Locomotion and Movement, Neural Control and Coordination

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

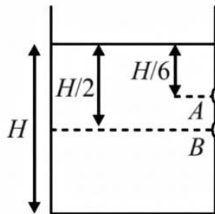
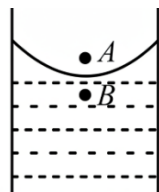
1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

- In a thermodynamic process, 200 J of heat is removed from an ideal monoatomic gas. If the work done by the gas is 50 J, the change in internal energy of the gas is:
 (1) +250 J (2) -250 J
 (3) -150 J (4) +150 J
- The mean free path of gas molecules of a gas is proportional to n^{th} power of diameter of molecules where n is equal to:
 (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$
 (3) -2 (4) 2
- The efficiency of a Carnot engine is $\frac{100}{3}\%$. When the sink temperature is reduced by 60 K keeping the source temperature fixed, the efficiency becomes 50%. Find the source temperature.
 (1) 300 K (2) 360 K
 (3) 420 K (4) 480 K
- At constant pressure, $P = 2 \text{ atm}$ ($1 \text{ atm} = 10^5 \text{ Pa}$), a monoatomic ideal gas expands from 1 L to 5 L. The heat given to the gas is:
 (1) 800 J (2) 1200 J
 (3) 2000 J (4) 3000 J
- Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In adiabatic compression of an ideal gas, the temperature of the gas increases.
Reason R: In an adiabatic process, $\Delta Q = 0$; hence work done on the gas increases its internal energy. In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.
- What is the dimensional formula of thermal conductivity?
 (1) $[ML^{-1}T^3\theta]$ (2) $[MLT^{-2}\theta^{-1}]$
 (3) $[ML^2T^{-2}\theta]$ (4) $[MLT^{-3}\theta^{-1}]$
- For a monoatomic ideal gas ($\gamma = 5/3$) undergoing a reversible adiabatic change, which relation is correct ($P \rightarrow$ pressure, $T \rightarrow$ temperature)?
 (1) $PT^{5/3} = \text{const}$ (2) $P^{-2/3}T^{5/3} = \text{const}$
 (3) $PT^{2/3} = \text{const}$ (4) $PT^{-5/2} = \text{const}$
- By what factor must the absolute temperature of a gas be multiplied so that its r.m.s. speed becomes 1.5 times the initial value?
 (1) 1.5 (2) 2.25
 (3) 1.25 (4) 3.0
- The range of liquid eject from orifice A is x times the range of liquid from orifice B as shown below. The value of x is:

 (1) 1 (2) $\frac{\sqrt{3}}{5}$
 (3) $\frac{\sqrt{5}}{3}$ (4) $\frac{3}{\sqrt{5}}$
- A gas mixture has 2 moles N_2 and 1 mole of He. Neglect vibrational modes. The total internal energy at absolute temperature T is: (R is universal gas constant)
 (1) $5RT$ (2) $\frac{13}{2}RT$
 (3) $8RT$ (4) $4RT$
- A mixture has 2 mol He and 1 mol N_2 . Neglect vibrational modes. The $\gamma = (C_p / C_v)$ of the mixture is:
 (1) $\frac{5}{3}$ (2) $\frac{7}{5}$
 (3) $\frac{17}{11}$ (4) $\frac{15}{11}$
- A capillary tube is immersed in a liquid, the shape of meniscus is shown below. Choose the **correct** statement.

 (1) Pressure at point A = Pressure at point B
 (2) Pressure at point A > Pressure at point B
 (3) Pressure at point A < Pressure at point B
 (4) Data insufficient to conclude

13. When temperature of a rod increases from T to $T + \Delta T$, its moment of inertia increases from I to $I + \Delta I$. If α is coefficient of linear expansion of rod, then $\frac{\Delta I}{I}$ is:

- (1) $2\alpha\Delta T$ (2) $\alpha\Delta T$
 (3) $\frac{\alpha\Delta T}{2}$ (4) $\frac{\alpha\Delta T}{3}$

14. Given below are two statements:

Statement I: The equation of continuity in fluid dynamics is based on conservation of mass.

Statement II: The Bernoulli's equation in fluid dynamics is applicable only for non-viscous incompressible steady flow.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

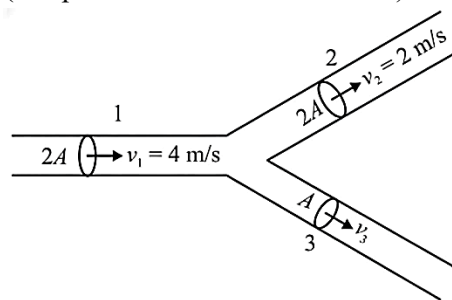
- (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.
15. Two hot bodies A and B of same surface area have absolute temperature in the ratio $2 : 1$ respectively. If their emissivity is in the ratio of $1 : 4$, then rate of heat emission will be in the ratio:
 (1) $1 : 8$ (2) $4 : 1$
 (3) $1 : 1$ (4) $1 : 2$
16. In an isothermal expansion of an ideal gas, the change in internal energy and work done by the gas are respectively:
 (1) negative, negative
 (2) zero, positive
 (3) zero, negative
 (4) positive, negative
17. Heat required to convert 10 g of ice at -10°C to water at 20°C at 1 atm pressure is:
 (1) 250 cal (2) 2916 cal
 (3) 850 cal (4) 1050 cal
18. A container filled with hot water cools from 60°C to 50°C in 4 minutes and 40°C to 30°C in 8 minutes. The approximate temperature of surrounding is:
 (1) 10°C (2) 15°C
 (3) 20°C (4) 30°C
19. n identical spherical drops each of radius r coalesce to form one drop (surface tension = σ). The energy released in the process:
 (1) $4\pi\sigma r^2(n^{2/3} - n)$ (2) $4\pi\sigma r^2(n - n^{2/3})$
 (3) $8\pi\sigma r^2(n - n^{2/3})$ (4) $4\pi\sigma r(n - n^{1/3})$

20. At constant water density, the absolute pressure at the bottom of a tank is 6 times the absolute pressure at a depth of 1 m from the top. If atmospheric pressure equals a 10 m water column, find the total depth H of the tank.

- (1) 56 m (2) 46 m
 (3) 60 m (4) 50 m

21. An incompressible liquid flows through a horizontal tube as shown in the following figure. The speed (v_3) of the fluid in tube 3 is:

(A represents area of cross-section)



- (1) 3 m/s (2) 4 m/s
 (3) 7 m/s (4) 1 m/s

22. In Maxwell's speed distribution curve on decreasing temperature, the number of gas molecules having most probable speed will:

- (1) decrease.
 (2) increase.
 (3) may increase or decrease.
 (4) remain constant.

23. Which postulate of kinetic theory justify why the internal energy of an ideal gas depends only on temperature?

- (1) Molecules have the same mass.
 (2) Molecular size is negligible relative to container volume.
 (3) There is no intermolecular potential energy except during instant collisions, so energy is purely kinetic.
 (4) Collisions with the walls are elastic.

24. A faulty thermometer reads 5°C as melting point of ice and 99°C as steam point. The correct temperature in $^\circ\text{F}$ when this faulty thermometer reads 52°C is:

- (1) 142°F (2) 130°F
 (3) 117°F (4) 122°F

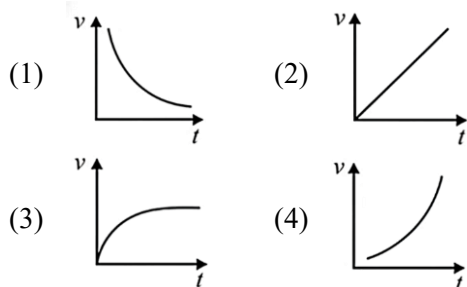
25. Two air bubbles in water have radii in the ratio $2 : 3$. What is the ratio of excess pressures inside them?

- (1) $2 : 3$ (2) $3 : 2$
 (3) $4 : 9$ (4) $9 : 4$

26. A black body emits radiations of maximum intensity for the wavelength of $4000 \text{ \AA}$ when the temperature of the body is 2227°C . If the temperature of the body is increased by 1000°C , then maximum intensity would be observed for wavelength of:

- (1) $2857 \text{ \AA}$ (2) $2560 \text{ \AA}$
(3) $2351 \text{ \AA}$ (4) $3248 \text{ \AA}$

27. A spherical ball is dropped in a deep viscous liquid. Which among the following curve correctly represents the speed of ball (v) as a function of time (t)?

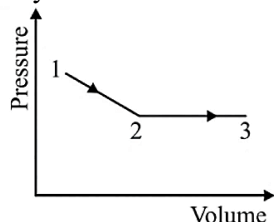


28. Water falls from 420 m height. What is the rise in temperature if all of its lost gravitational potential energy is converted to thermal energy?

(Given: $g = 10 \text{ m s}^{-2}$, $C_w = 4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$)

- (1) $0.42 ^\circ\text{C}$ (2) $0.50 ^\circ\text{C}$
(3) $1.00 ^\circ\text{C}$ (4) $1.20 ^\circ\text{C}$

29. Consider the thermodynamic processes ($1 \rightarrow 2$ and $2 \rightarrow 3$) on a system in figure below. The work done by the system:



- (1) Continuously increases
(2) Continuously decreases
(3) First increases then decreases
(4) First decreases then increases

30. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Viscosity of liquids decreases with rise in temperature.

Reason R: Cohesive forces in liquid increases with a rise in temperature.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false
(2) A is false but R is true
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

31. Liquids A and B are at 40°C and 30°C respectively. When mixed in equal masses, the mixture temperature is 35°C . What is the ratio of specific heats of liquids A and B, ($c_A : c_B$)?

- (1) $1 : 1$ (2) $3 : 2$
(3) $2 : 3$ (4) $4 : 3$

32. A monatomic ideal gas is considered. Which statements about its process molar specific heats C are **correct**?

A. Isobaric: $C = \frac{5R}{2}$

B. Isochoric: $C = \frac{3R}{2}$

C. Adiabatic: $C = 0$

D. Isothermal: $C = \infty$

- (1) A, B, C only (2) B, C, D only
(3) A, C, D only (4) A, B, C and D

33. A slab consist of two rods of different materials of same thickness and equal length joined end to end and having the thermal conductivities K and $2K$ respectively. The equivalent thermal conductivity of the slab is:

- (1) $3K$ (2) $\frac{3K}{2}$
(3) $\frac{2K}{3}$ (4) $\frac{4K}{3}$

34. The ratio of $v_{rms} : v_{mp} : v_{avg}$ for ideal gas is (symbols have their usual meaning)?

- (1) $\sqrt{3} : \sqrt{2} : \sqrt{\frac{\pi}{8}}$ (2) $\sqrt{3} : \sqrt{2} : \sqrt{\frac{8}{3}}$
(3) $\sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2}$ (4) $\sqrt{3} : \sqrt{2} : \sqrt{\frac{8}{\pi}}$

35. What is the work done to expand a soap bubble from radius $R = 0.10 \text{ m}$ to 0.20 m . If surface tension

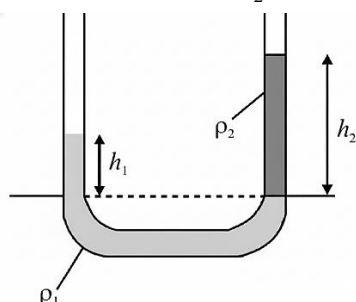
$T = 0.50 \text{ N m}^{-1}$?

- (1) $(0.24\pi) \text{ J}$ (2) $(0.12\pi) \text{ J}$
(3) $(0.06\pi) \text{ J}$ (4) $(0.48\pi) \text{ J}$

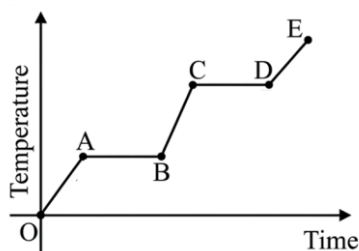
36. The value of γ (adiabatic index) is maximum for which of the following gas at room temperature?

- (1) Monoatomic gas
(2) Diatomic gas
(3) Non-linear triatomic gas
(4) Linear-triatomic gas

37. Two immiscible liquids are filled in a U-tube having densities ρ_1 and ρ_2 . The ratio of height of the liquids above their interface $\frac{h_1}{h_2}$ as shown in figure is:



- (1) $\frac{\rho_1}{\rho_2}$ (2) $\frac{\rho_2}{\rho_1}$
 (3) $\left(\frac{\rho_1}{\rho_2}\right)^2$ (4) $\left(\frac{\rho_2}{\rho_1}\right)^2$
38. If a body cools from 70°C to 50°C in 10 min. Surrounding temperature is 30°C . What will its temperature after the next 10 min?
 (1) 35°C (2) 38°C
 (3) 40°C (4) 42°C
39. The maximum efficiency of Carnot engine is 60%. If the temperature of source reservoir is 227°C , then the temperature of sink reservoir is:
 (1) 91K (2) 91°C
 (3) -73K (4) -73°C
40. To a given mass of solid, heat is supplied at constant rate and a graph is plotted between temperature and time as shown below.



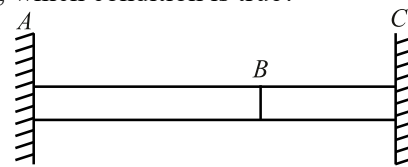
Match the phenomenon given in List-I to the corresponding region given in List-II.

List-I		List-II	
(A)	Heating of solid	(I)	CD
(B)	Melting of solid	(II)	BC
(C)	Heating of liquid	(III)	OA
(D)	Boiling of liquid	(IV)	AB

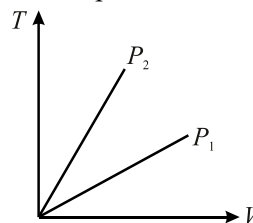
Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
 (2) A-IV, B-III, C-II, D-I
 (3) A-III, B-IV, C-I, D-II
 (4) A-III, B-IV, C-II, D-I

41. Two equal-area rods with Young's moduli Y_1, Y_2 and linear expansion coefficients α_1, α_2 are fixed between rigid walls and joined at B. For no movement of point B for any temperature change ΔT , which condition is true?



- (1) $\alpha_1 = \alpha_2$ (2) $Y_1\alpha_1 = Y_2\alpha_2$
 (3) $\frac{Y_1}{Y_2} = \frac{\alpha_1}{\alpha_2}$ (4) Both (1) and (2)
42. The **correct** kelvin-Planck statement which is related to second law of thermodynamics is:
 (1) It is possible to construct a device whose purpose is to convert the entire heat energy absorbed from the source into an equal amount of work.
 (2) It is possible to construct a device which can transfer the heat from a lower temperature reservoir to higher temperature reservoir without any external work input.
 (3) It is impossible to construct a device which can transfer the heat from a lower temperature reservoir to higher temperature reservoir without any external work input.
 (4) It is impossible to construct a device whose purpose is to convert the entire heat energy absorbed from the source into an equal amount of work.
43. A small hole is made at some depth so that the ejected liquid has maximum horizontal range. The free surface is at height $H = 2.5\text{ m}$ above the ground and $g = 10\text{ m s}^{-2}$. What time does a fluid element from the hole take to reach the ground?
 (1) 0.40 s (2) 0.50 s
 (3) 0.60 s (4) 0.80 s
44. When a capillary tube of radius 0.3 mm is dipped in a liquid, a mass $M = 6\text{ gram}$ rises. If another tube of radius 0.9 mm is used, the mass that rises is
 (1) 2 gram (2) 6 gram
 (3) 12 gram (4) 18 gram
45. Temperature-Volume (T - V) graph of an ideal gas for two different isobaric processes is shown in figure. If P_1 and P_2 are the pressure of the gas then,



- (1) $P_1 < P_2$
 (2) $P_1 = P_2$
 (3) $P_1 > P_2$
 (4) depends on atomicity of gas.

46. How can a galvanic cell be converted into an electrolytic cell?

- (1) Applying an external opposite potential greater than E_{cell}° .
- (2) Reversing the flow of ions in the salt bridge.
- (3) Applying an external opposite potential smaller than E_{cell}° .
- (4) Exchanging the electrodes at anode and cathode.

47. The number of **correct** statements is:

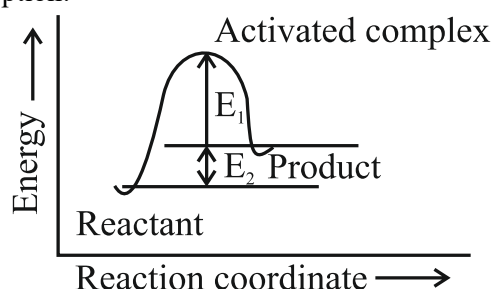
- A. E_{cell} is an intensive property.
- B. A negative standard reduction potential for a couple means its reduced member is a stronger reducing agent than H_2/H^+ .
- C. The electricity needed for a given chemical change at an electrode depends on the stoichiometric electron count in the electrode reaction.
- D. The amount of chemical change during electrolysis is proportional to the total charge passed through the electrolyte.

- (1) 1
- (2) 2
- (3) 3
- (4) 4

48. Integrated rate law equation for a first order gas phase reaction $\text{A}(\text{g}) \rightarrow \text{B}(\text{g}) + \text{C}(\text{g})$ is given by; (where P_i is initial pressure and P_t is total pressure at time t)

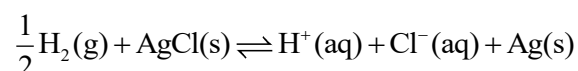
- (1) $k = \frac{2.303}{t} \times \log \frac{P_i}{(2P_i - P_t)}$
- (2) $k = \frac{2.303}{t} \times \log \frac{2P_i}{(2P_i - P_t)}$
- (3) $k = \frac{2.303}{t} \times \log \frac{(2P_i - P_t)}{P_i}$
- (4) $k = \frac{2.303}{t} \times \log \frac{3P_i}{(2P_i - P_t)}$

49. Consider the given figure and choose the **correct** option:



- (1) Activation energy of both forward and backward reaction is $E_1 + E_2$ and reactant is more stable than product.
- (2) Activation energy of forward reaction is $E_1 + E_2$ and product is more stable than reactant.
- (3) Activation energy of backward reaction is E_1 and product is more stable than reactant.
- (4) Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant.

50. The reaction



occurs in which of the following galvanic cells?

- (1) $\text{Pt} | \text{H}_2(\text{g}) | \text{KCl}(\text{aq}) | \text{AgCl}(\text{s}) | \text{Ag}(\text{s})$
- (2) $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl}(\text{aq}) | \text{AgCl}(\text{s}) | \text{Ag}(\text{s})$
- (3) $\text{Ag} | \text{AgCl}(\text{s}) | \text{KCl}(\text{aq}) | \text{AgCl}(\text{s}) | \text{Ag}(\text{s})$
- (4) $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl}(\text{aq}) | \text{AgNO}_3(\text{aq}) | \text{Ag}(\text{s})$

51. For the cell $\text{Pt} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{Ag}^+(\text{aq}) | \text{Ag}(\text{s})$, the standard emf is $E_{\text{cell}}^{\circ} = +0.50 \text{ V}$. For the overall reaction written with electrons cancelled, the value of ΔG° is (in kJ mol^{-1})

$$(F = 9.65 \times 10^4 \text{ C mol}^{-1})$$

- (1) -48.8
- (2) -65.5
- (3) -96.5
- (4) -120

52. Based on the data given below:

$$E_{\text{Ce}^{4+}/\text{Ce}^{3+}}^{\circ} = +1.70 \text{ V}, E_{\text{Cl}_2/\text{Cl}^-}^{\circ} = +1.36 \text{ V},$$

$$E_{\text{MnO}_4^-/\text{Mn}^{3+}}^{\circ} = +1.49 \text{ V}, E_{\text{Fe}^{3+}/\text{Fe}}^{\circ} = -0.44 \text{ V}.$$

The strongest reducing agent among the following is:

- (1) MnO_4^-
- (2) Cl^-
- (3) $\text{Fe}(\text{s})$
- (4) Ce^{3+}

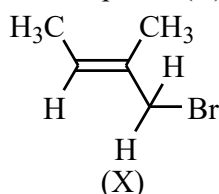
53. If the hydrogen-ion concentration of a solution increases by a factor of 100, the pH of the solution will:

- (1) increase by 2 units
- (2) decrease by 2 units
- (3) increase by 100 units
- (4) decrease by 100 units

54. 500 mL of 0.020 M HCl is mixed with 500 mL of 0.010 M H_2SO_4 . Assuming complete dissociation, the pH of the mixture is (use $\log 2 = 0.30$).

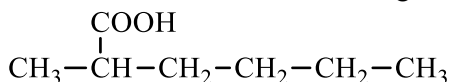
- (1) 1.40
- (2) 1.60
- (3) 1.70
- (4) 1.90

55. Which of the following is the **correct** IUPAC name of given organic compound (X)?

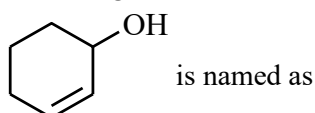


- (1) 2-Bromo-2-methylbut-2-ene
(2) 3-Bromo-3-methylprop-2-ene
(3) 1-Bromo-2-methylbut-2-ene
(4) 4-Bromo-3-methylbut-2-ene

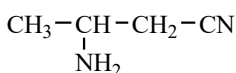
56. The IUPAC name of the following compound is:



- (1) 2-Methylhexanoic acid.
(2) 2-Methylcarbonylhexane.
(3) 2-Methylhexanoate.
(4) 5-Methylhexanoic acid.
57. According to IUPAC system, the compound



- (1) Cyclohex-1-en-2-ol
(2) 1-Hydroxyhex-2-ene
(3) Cyclohex-1-en-3-ol
(4) Cyclohex-2-en-1-ol
58. IUPAC name of following compound is:



- (1) 2-Aminopentanenitrile
(2) 2-Aminobutanenitrile
(3) 3-Aminobutanenitrile
(4) 3-Aminopropanenitrile
59. Given below are two statements:
- Statement I:** IUPAC name of $\text{HO}-\text{CH}_2-(\text{CH}_2)_3-\text{CH}_2-\text{COCH}_3$ is 7-hydroxyheptan-2-one.

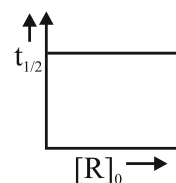
Statement II: 2-oxoheptan-7-ol is the correct IUPAC name for above compound.

In the light of the above statements. Choose the *most appropriate* answer from the options given below.

- (1) Statement-I is correct but Statement-II is incorrect.
(2) Both Statement-I and Statement-II are incorrect.
(3) Both Statement-I and Statement-II are correct.
(4) Statement-I is incorrect but Statement-II is correct.

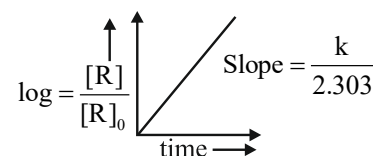
60. Given below are two statements:

Statement I:



is valid for first order reaction.

Statement II:



is valid for first order reaction.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are true
- 61.

Which of the following is the **correct** IUPAC name of the above compound?

- (1) p-acetamido chlorocyclohexane.
(2) 4-chloro-N-methylcyclohexane carboxamide
(3) N-methyl amido chlorocyclohexane
(4) None of these
62. Which of the following is the condensed formula for 3-oxopentanal?
- (1) $(\text{CH}_3)_2\text{CHCOCHO}$
(2) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CHO}$
(3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCHO}$
(4) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CHO}$

63. For the reaction of H_2 with I_2 , the rate constant is $2.5 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 327°C and $1.0 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 527°C . The activation energy for the reaction, in kJ mol^{-1} is ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$) ($\log 5 = 0.69$, $\log 2 = 0.30$)

- (1) 150 (2) 59.3
(3) 72.8 (4) 165.5

64. Which of the following is an example of pseudo unimolecular reaction?

- (1) $\text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOH} + \text{CH}_3\text{OH}$
(2) $\text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O} \xrightarrow{\text{OH}^-} \text{CH}_3\text{COOH} + \text{CH}_3\text{OH}$
(3) $2\text{FeCl}_3 + \text{SnCl}_2 \rightarrow \text{SnCl}_4 + 2\text{FeCl}_2$
(4) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

65. Match the **List-I** with **List-II**.

List-I (General formula)		List-II (Class of compounds)	
(A)	C_nH_{2n}	(I)	Alkyne
(B)	$C_nH_{2n+2}O$	(II)	Alcohol
(C)	C_nH_{2n-2}	(III)	Aldehyde
(D)	$C_nH_{2n}O$	(IV)	Alkene

Choose the **correct** answer from the options given below;

- (1) A-IV, B-II, C-I, D-III
 (2) A-I, B-II, C-III, D-IV
 (3) A-IV, B-III, C-I, D-II
 (4) A-III, B-IV, C-II, D-I

66. The IUPAC name of $\begin{array}{c} CH_2-CH-CH_2 \\ | \quad | \quad | \\ CN \quad CN \quad CN \end{array}$ is:

- (1) Propane-1,2,3-tricarbonitrile
 (2) 3-Cyanopentane-1,5-dinitrile
 (3) 3,4-Dicyanobutanenitrile
 (4) 1,2-Dicyanobutanenitrile

67. Which of the following compounds **cannot** act as a Lewis base?

- (1) IF_3 (2) PF_5
 (3) NH_3 (4) $SOCl_2$

68. The pH of pure water is 7.0 at 25 °C. If the water is heated to 60 °C, its pH will:

- (1) remain the same (2) decrease
 (3) increase (4) first increase then decrease

69. The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-31} . The solubility of Cr^{3+} ions in a saturated solution of $Cr(OH)_3$ will be:

- (1) $(18 \times 10^{-31})^{1/4}$ (2) $(4.86 \times 10^{-29})^{1/4}$
 (3) $(18 \times 10^{-31})^{1/2}$ (4) $(2.22 \times 10^{-32})^{1/4}$

70. pH of 10^{-8} M HNO_3 at 25°C is:

- (1) 6.98 (2) 7.98
 (3) 5.98 (4) 8.02

71. Given $E^\circ_{Cu^{2+}/Cu} = +0.32$ V and $E^\circ_{Cu^+/Cu} = +0.48$ V, the value of $E^\circ_{Cu^{2+}/Cu^+}$ is:

- (1) -0.16 V (2) +0.16 V
 (3) -0.20 V (4) +0.20 V

72. The emf of the cell

$X|X^{2+}(0.010\text{ M})||Y^{2+}(0.10\text{ M})|Y$ at 298 K is:

(Given: $E^\circ_{X^{2+}/X} = -0.40$ V, $E^\circ_{Y^{2+}/Y} = +0.40$ V, and $\frac{2.303 RT}{F} = 0.06$ V.)

- (1) 0.83 V (2) 0.67 V
 (3) 0.57 V (4) 0.47 V

73. 40 mL of 0.10 M acetic acid is titrated with 0.10 M NaOH. After adding 20 mL base, the pH is [take $pK_a(CH_3COOH) = 4.76$].

- (1) 3.76 (2) 4.76
 (3) 5.76 (4) 7.00

74. Arrange the following 0.020 M solutions in decreasing order of pOH:

- A. HCl B. NaOH
 C. NH_4Cl D. NaCl
 (1) $B > D > C > A$
 (2) $A > C > D > B$
 (3) $C > D > A > B$
 (4) $D > C > B > A$

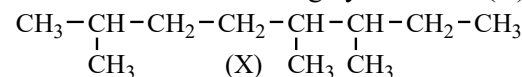
75. 25 mL of 0.10 M HCl is added to 25 mL of 0.20 M NH_4OH . The pH of the resulting mixture is ($pK_b(NH_4OH) = 4.7$).

- (1) 5.2 (2) 9.3
 (3) 5.0 (4) 9.7

76. The pH of 0.040 M NH_4Cl solution is ($K_b(NH_4OH) = 1.0 \times 10^{-6}$), ($\log 2 = 0.3$).

- (1) 5.05 (2) 4.7
 (3) 5.35 (4) 4.80

77. IUPAC name of following hydrocarbon (X) is:



- (1) 2-Ethyl-3,6-dimethylheptane
 (2) 2-Ethyl-2,6-diethylheptane
 (3) 2,5,6-Trimethyloctane
 (4) 3,4,7-Trimethyloctane

78. Identify the factor that does **not** affect the electrolytic conductance of a solution (at fixed temperature).

- (1) Nature of the electrolyte added
 (2) Nature of the electrode used
 (3) Concentration of the electrolyte
 (4) Nature of the solvent used

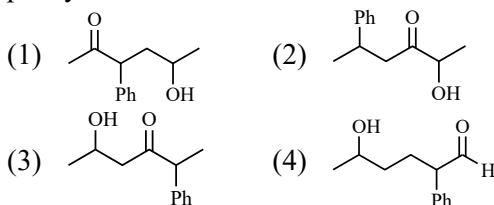
79. For the lead storage battery, pick the **correct** statements:

- A. During charging, $PbSO_4$ on the anode is converted back to $Pb(s)$.
 B. During charging, $PbSO_4$ on the cathode is converted back to PbO_2 .
 C. The battery consists of a copper grid packed with PbO_2 as the cathode and spongy Pb as the anode.

Choose the **correct** answer:

- (1) A only (2) B only
 (3) A and B only (4) C only

80. What is the **correct** structure for 5-hydroxy-2-phenyl hexan-3-one?



81. The **correct** statements are:

- A. At a fixed temperature, larger activation energy implies a smaller rate constant.
 B. A higher activation energy gives a larger temperature coefficient (greater sensitivity of k to T).
 C. A given rise in temperature causes a larger fractional increase in k at lower temperatures than at higher ones.
 D. A plot of $\ln k$ vs $1/T$ is a straight line with slope $+E_a/R$.

- (1) A, B only (2) A, B and C only
 (3) B and D only (4) C and D only

82. The reaction that occurs at cathode in H_2-O_2 fuel cell is:

- (1) $H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$
 (2) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
 (3) $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$
 (4) $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$

83. A first-order reaction has rate constant $k = 3.0 \times 10^{-3} s^{-1}$. Which of the following statements are **correct**? ($\log 10 = 1$, $\ln 2 = 0.693$)

- A. The half-life is $0.693/k$ is 231 s.
 B. The rate and the rate constant have the same unit.
 C. The reaction finishes exactly in five half-lives.
 (1) A only (2) B only
 (3) C only (4) A and B only

84. In the cell commonly used in transistors, the reaction at the cathode involves

- (1) reduction of Mn from +4 to +3
 (2) oxidation of Mn from +3 to +4
 (3) reduction of Mn from +7 to +2
 (4) oxidation of Mn from +2 to +7

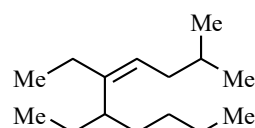
85. For the reaction $3A + 2B + \frac{1}{2}C \rightarrow 4P$, which relation is **correct**?

- (1) $-\frac{1}{3} \frac{d[A]}{dt} = \frac{1}{4} \frac{d[B]}{dt} = -\frac{1}{2} \frac{d[P]}{dt} = -2 \frac{d[C]}{dt}$
 (2) $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = -\frac{d[C]}{dt} = \frac{d[P]}{dt}$
 (3) $-\frac{1}{3} \frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt} = -2 \frac{d[C]}{dt} = \frac{1}{4} \frac{d[P]}{dt}$
 (4) $-3 \frac{d[A]}{dt} = -2 \frac{d[B]}{dt} = -\frac{1}{2} \frac{d[C]}{dt} = 4 \frac{d[P]}{dt}$

86. For $A_2 + B_2 \rightleftharpoons 2AB$, the activation energies for the forward and backward reactions are 150 and 170 kJ mol^{-1} , respectively. A catalyst lowers both activation energies by 50 kJ mol^{-1} . Which statement is **correct**?

- (1) The standard Gibbs energy change is altered by the catalyst.
 (2) The enthalpy change remains -20 kJ mol^{-1} for both catalysed and uncatalysed paths.
 (3) The catalyst can make a non-spontaneous reaction spontaneous.
 (4) The enthalpy change for the catalysed reaction differs from that of the uncatalysed reaction.

87. The IUPAC name of the given compound is:



- (1) 2,4,5-Triethylnon-3-ene
 (2) 5,6-Diethyl-2-methyldec-4-ene
 (3) 2,4,5-Triethyloct-3-ene
 (4) 3-Ethyl-5-methylhept-3-ene

88. For a strong electrolyte, a plot of molar conductivity Λ_m against $\sqrt{c}$ is a straight line with a negative slope. If c is in mol L^{-1} and Λ_m in $\text{S cm}^2 \text{ mol}^{-1}$, the **correct** unit of the slope is:

- (1) $\text{S cm}^2 \text{ mol}^{-3/2} \text{ L}^{1/2}$
 (2) $\text{S cm}^2 \text{ mol}^{-1} \text{ L}^{1/2}$
 (3) $\text{S cm}^2 \text{ mol}^{-1/2} \text{ L}^{-1/2}$
 (4) $\text{S cm}^2 \text{ mol}^{-3/2} \text{ L}^{-1/2}$

89. At 975 K the following data were recorded for $2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$

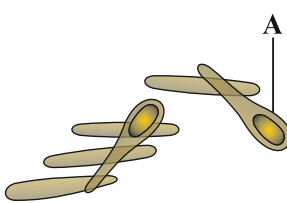
Expt.	$[NO](\text{mol L}^{-1})$	$[H_2](\text{mol L}^{-1})$	Rate ($\text{mol L}^{-1} \text{ s}^{-1}$)
1	1.0×10^{-4}	2.0×10^{-4}	4.0×10^{-8}
2	2.0×10^{-4}	2.0×10^{-4}	1.6×10^{-7}
3	2.0×10^{-4}	4.0×10^{-4}	3.2×10^{-7}

The order with respect to NO is _____.

- (1) 0
 (2) 1
 (3) 2
 (4) 3

90. Which electrolyte is used to obtain $H_2S_2O_8$ by electrolysis?

- (1) Dilute Na_2SO_4 solution
 (2) Cold, concentrated H_2SO_4
 (3) Dilute H_2SO_4
 (4) Acidified dilute Na_2SO_4

91. Find the **correctly** matched pair.
- (1) Diphtheria – Kusht rog
 - (2) TMV – Appears rod-shaped when magnified about 1,00,000–1,50,000x
 - (3) Leprosy – Gal ghotu
 - (4) Influenza – Bacterial disease
92. Which of the following is smallest organism/structure among the following?
- (1) *Saccharomyces*
 - (2) *Mycoplasma*
 - (3) Prions
 - (4) Viroids
93. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Bottled fruit juices bought from the market are clearer as compared to those made at home.
Reason R: The bottled juices are clarified by the use of lipases.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
94. Identify the given organism and the part labelled as 'A'.
- 
- (1) *Anabaena* – heterocyst
 - (2) *Bacillus* – spore
 - (3) Diatoms – cell wall
 - (4) Yeast – nucleus
95. Which of the following sets of spores are sexual spores in fungi?
- (1) oospores and ascospores
 - (2) zygospores and aplanospores
 - (3) zoospores and conidia
 - (4) sporangiospores and basidiospores
96. A traditional drink of some parts of southern India, toddy is made by fermentation of:
- (1) sap from palms.
 - (2) bamboo shoots.
 - (3) malted cereals and fruit juices.
 - (4) cellulosic material.

97. Which of the following is a **correct** feature of lichens?
- (1) The mycobiont component provides food for its partner.
 - (2) The phycobiont component provides shelter and absorbs mineral nutrients and water for its partner.
 - (3) They improve air quality by purifying the air and growing in polluted areas.
 - (4) They are not classified under the five-kingdom classification proposed by Whittaker.
98. Which of the following sets of organisms are photosynthetic and nitrogen-fixing?
- (1) *Anabaena* and *Oscillatoria*
 - (2) *Azospirillum* and *Azotobacter*
 - (3) *Rhizobium* and *Nostoc*
 - (4) *Glomus* and *Staphylococci*
99. The cavity that opens to the outside of the cell surface, into which water laden with food is directed, is known as:
- (1) pellicle.
 - (2) gullet.
 - (3) plasmodium.
 - (4) pseudopodia.
100. Identify the **correct** sequence of events that occur during sewage treatment in a sewage treatment plant.
- A. Sedimentation of flocs
 - B. Transfer of primary effluent to a large aeration tank
 - C. Transfer of activated sludge to an anaerobic sludge digester
 - D. Sequential filtration and sedimentation
- Choose the **correct** answer from the options given below:
- (1) B → D → C → A
 - (2) D → B → A → C
 - (3) D → A → B → C
 - (4) A → C → B → D
101. Given below are two statements:
- Statement I:** All protozoans are heterotrophs and live as predators or parasites.
- Statement II:** Protists have a well-defined nucleus and other membrane-bound organelles and all members possess a cellulosic cell wall.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

102. During anaerobic digestion of flocs in sewage treatment plant, which of the following combinations of gases are released?

- (1) CH₄, CO₂ and H₂S
- (2) CO₂, O₂ and NO₂
- (3) CO, H₂ and CH₄
- (4) O₂, CO₂ and H₂

103. Identify the **odd** one with respect to the type of genetic material usually present in the causative agent of the following diseases.

- (1) Sleeping sickness
- (2) Citrus canker
- (3) Potato spindle tuber
- (4) Rust in wheat

104. Which of the following statements **correctly** describes approach of organic farming?

- (1) Pests should be kept at manageable levels, as they help maintain the balance of the ecosystem.
- (2) Pests should be fully eradicated, as they are always harmful to the ecosystem.
- (3) Pests should be controlled through the use of non-harmful chemical pesticides.
- (4) Pests should be kept as such to ensure maximum diversity.

105. Which of the following features is **not** common between Ascomycetes and Deuteromycetes?

- (1) Type of spores produced asexually
- (2) Mycelium is septate and branched
- (3) Heterotrophic mode of nutrition
- (4) Sexual spores are produced endogenously

106. Which of the following set of products are generally produced by using fungus?

- (1) Idli dough and breads
- (2) Penicillin and statin
- (3) Cyclosporin A and Streptokinase
- (4) Butyric acid and acetic acid

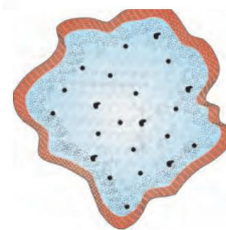
107. Which of the following are **correct** features of members of Kingdom Monera?

- A. They can also live in hot springs, deserts, snow, and deep oceans.
- B. Comma-shaped bacteria are called cocci.
- C. Bacterial structure and behaviour are very simple.
- D. They show the most extensive metabolic diversity.

Choose the **correct** answer from the options given below:

- (1) A, B, and C only
- (2) C and D only
- (3) A and B only
- (4) A and D only

108. The organism shown below:



- (1) are photosynthetic protists.
- (2) form a plasmodium under unfavourable conditions.
- (3) form a spore-bearing fruiting body under favourable conditions.
- (4) produce spores that have a true wall and disperse through air currents.

109. Which of the following organisms does **not** possess photosynthetic pigment?

- (1) Desmids
- (2) *Nostoc*
- (3) *Euglena*
- (4) *Paramecium*

110. An enzyme used as a 'clot buster' to remove clots from the blood vessels of patients who have suffered a myocardial infarction is obtained through genetic engineering from a:

- (1) bacterium, *Staphylococcus*.
- (2) bacterium, *Streptococcus*.
- (3) fungus, *Trichoderma polysporum*.
- (4) fungus, *Saccharomyces cerevisiae*.

111. In which of the following organisms is the cell wall made up of polysaccharides and amino acids?

- (1) *Aspergillus*
- (2) Bladderwort
- (3) *Nostoc*
- (4) Desmids

112. Given below are two statements:

Statement I: Baculoviruses are pathogens that attack insects and other arthropods.

Statement II: *Nucleopolyhedrovirus* shows no negative impacts on plants, mammals, birds, fish or even on non-target insects.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

113. Which of the following is **not** associated with fungi?

- (1) White spots on mustard leaves
- (2) Some are used to make bread and beer
- (3) Some are the source of antibiotics
- (4) Some are pathogenic and causes disease such as citrus canker

114. Match List I with List II.

List-I		List II	
(A)	<i>Trichoderma</i> sp.	(I)	Used to get rid of aphids
(B)	Ladybird	(II)	Effective biocontrol agent of several plant pathogens
(C)	<i>Bacillus thuringiensis</i>	(III)	Free-living nitrogen fixer
(D)	<i>Azospirillum</i>	(IV)	Releases toxin in the gut of larvae of butterfly caterpillars to kill it

Choose the *most appropriate* answer from the options given below:

- (1) A–I, B–III, C–II, D–IV
- (2) A–III, B–II, C–I, D–IV
- (3) A–II, B–I, C–IV, D–III
- (4) A–IV, B–I, C–III, D–II

115. Which of the following pairs of organisms or groups of organisms were not placed together in the classification given by Carolus Linnaeus?

- (1) *Spirogyra* and *Chlamydomonas*
- (2) *Chlorella* and *Paramecium*
- (3) Fungi and photosynthetic bacteria
- (4) Mosses and blue-green algae

116. Match List-I with List-II.

List-I		List-II	
(A)	<i>Saccharomyces</i>	(I)	Protozoan
(B)	<i>Trichoderma</i>	(II)	Ascomycetes
(C)	<i>Ustilago</i>	(III)	Basidiomycetes
(D)	<i>Trypanosoma</i>	(IV)	Deuteromycetes

Choose the **correct** answer from the options given below:

- (1) A–IV, B–III, C–II, D–I
- (2) A–III, B–II, C–I, D–IV
- (3) A–II, B–IV, C–III, D–I
- (4) A–I, B–II, C–III, D–IV

117. Which scientist demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants and named the new pathogen “virus”?

- (1) Dmitri Ivanowsky
- (2) M.W. Beijerinck
- (3) W.M. Stanley
- (4) T.O. Diener

118. Which of the following statements about biochemical oxygen demand (BOD) is **incorrect?**

- (1) It is the amount of oxygen that would be consumed if all the organic matter in one litre of water were oxidized by bacteria.
- (2) The greater the BOD of wastewater, the higher is its polluting potential.
- (3) It is higher in water released from the sewage treatment plant as compared to the primary effluent.
- (4) The higher the organic matter in water, the higher will be the BOD.

119. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Plants are photosynthetic, but a few members are partially heterotrophic.

Reason R: Plants are chlorophyll-containing organisms, but some are insectivorous or parasitic.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

120. How many of the following are considered as microbes?

- A. Protozoa
- B. Bacteria
- C. Fungi
- D. Viruses
- E. Prions

Choose the **correct** answer from the options given below:

- (1) Two only
- (2) Three only
- (3) Four only
- (4) Five

121. The fungi causing rust in wheat:

- (1) has aseptate and coenocytic mycelium.
- (2) produce sexual spores endogenously in sac-like structures.
- (3) do not have sex organs.
- (4) produce conidia.

122. The flocs:

- (1) are formed as a result of sequential filtration.
- (2) are masses of anaerobic bacteria associated with fungal filaments forming mesh like structures.
- (3) are formed as a result of the growth of microbes in large aeration tanks.
- (4) are the settled material after sedimentation during primary treatment.

123. The members of fungi that have hyphae as continuous tubes filled with multinucleated cytoplasm:

- (1) are found in aquatic habitats and on decaying wood in moist and damp places, or as obligate parasites on plants.
- (2) reproduce by zoospores or conidia formed endogenously.
- (3) have gametes that are always isogamous.
- (4) include morels and truffles.

124. Microbes that are used commercially to produce alcohol:

- (1) are multicellular fungi.
- (2) perform fermentation under aerobic conditions.
- (3) are grown in very large vessels called fermenters for alcohol production.
- (4) belong to the kingdom that includes all eukaryotic unicellular organisms.

125. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Thermoacidophiles	(I)	Mucilaginous sheath
(B)	<i>Nostoc</i>	(II)	Play a great role in recycling nutrients
(C)	Heterotrophic bacteria	(III)	Found in hot springs
(D)	Chemoautotrophic bacteria	(IV)	Most abundant in nature

Choose the *most appropriate* answer from the options given below:

- (1) A–III, B–I, C–IV, D–II
- (2) A–IV, B–I, C–III, D–II
- (3) A–III, B–I, C–II, D–IV
- (4) A–IV, B–III, C–I, D–II

126. During curd formation, LAB produce A which coagulates and B the milk proteins.

- (1) A – lactic acid, B – completely digests
- (2) A – vitamin B₁₂, B – partially digests
- (3) A – lactic acid, B – partially digests
- (4) A – vitamin B₁₂, B – completely digests

127. Dikaryophase in fungi is seen in:

- (1) Deuteromycetes and Basidiomycetes.
- (2) Ascomycetes and Phycomycetes.
- (3) Ascomycetes and Basidiomycetes.
- (4) Phycomycetes and Deuteromycetes.

128. Which of the following alcoholic beverages are produced without involvement of distillation process?

- (1) Whisky and beer
- (2) Brandy and rum
- (3) Wine and beer
- (4) Rum and wine

129. How many of the following diseases are caused by organisms or structures that are **not** included in Whittaker's five-kingdom classification?

- A. Tetanus
- B. Cholera
- C. Mumps
- D. Herpes
- E. Smut

Choose the **correct** answer from the options given below:

- (1) Two only
- (2) Three only
- (3) Four only
- (4) Five

130. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Penicillium notatum</i>	(I)	Production of Swiss cheese
(B)	<i>Aspergillus niger</i>	(II)	Used for the production of citric acid
(C)	<i>Saccharomyces cerevisiae</i>	(III)	Also known as baker's and brewer's yeast
(D)	<i>Propionibacterium sharmanii</i>	(IV)	Production of antibiotic

Choose the **correct** answer from the options given below:

- (1) A–IV, B–II, C–I, D–III
- (2) A–II, B–IV, C–I, D–III
- (3) A–IV, B–II, C–III, D–I
- (4) A–I, B–III, C–II, D–IV

131. Which of the following is **correct** about virus?

- (1) A virus is a nucleoprotein and the protein is infectious.
- (2) Viruses that infect animals have either single or double stranded RNA or double stranded DNA.
- (3) Viruses could be crystallised and crystals consist largely of nucleic acid.
- (4) Bacterial virus usually has single stranded RNA as genetic material.

132. The bioactive molecule obtained from *Monascus purpureus*:

- (1) is used as an immunosuppressive agent in organ transplant patients.
- (2) acts by competitively inhibiting the enzyme responsible for the synthesis of cholesterol.
- (3) provides resistance against plant pathogens.
- (4) is used in the formulation of detergents.

133. In the production of which product there is no evolution of CO₂?

- (1) Dough used for making dosa
- (2) Swiss cheese production
- (3) *Methanobacterium* growing on cellulosic material
- (4) Curd formation

134. Which of the following benefits is not generally observed in the associated plant from the fungal species in a mycorrhizal association?

- (1) Fixation of atmospheric nitrogen
- (2) Tolerance to salinity and drought
- (3) Overall increase in plant growth and development
- (4) Resistance to root-borne pathogens

135. Which of the following statements is **correct** about biogas?

- (1) Biogas is a mixture of gases containing equal proportions of methane and CO₂.
- (2) Apart from methane, the other gases in biogas may vary depending on the type of raw material used during its production.
- (3) Bacteria producing biogas grow aerobically on cellulosic material.
- (4) It is also produced in the rumen of all animals.

136. Which of the following structures in a neuron contains Nissl's granules?

- (1) Axon
- (2) Synaptic knob
- (3) Cell body
- (4) Neurotransmitters

137. In an experiment, the midbrain of an animal is surgically damaged. Which of the following abilities will **most** likely be affected?

- (1) Regulation of heartbeat and breathing.
- (2) Integration of visual and auditory signals for orientation.
- (3) Maintenance of body temperature.
- (4) Coordination of voluntary muscle movement.

138. Which of the following characteristics is **not** associated with white muscle fibres?

- (1) Less quantity of myoglobin.
- (2) Number of mitochondria are few.
- (3) Amount of sarcoplasmic reticulum is low.
- (4) Depend on anaerobic process for energy.

139. Striated muscles are primarily involved in;

- A. Locomotory actions.
- B. Changes in body postures.
- C. Transportation of food through digestive tract.
- D. Transportation of gametes through genital tract.

- (1) A and B only
- (2) B and C only
- (3) C and D only
- (4) A and D only

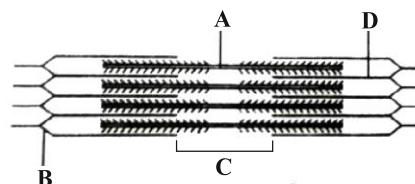
140. Match List-I with List-II.

List-I		List-II	
(A)	Tetany	(I)	Progressive degeneration of skeletal muscles
(B)	Osteoporosis	(II)	Auto immune disorder
(C)	Muscular dystrophy	(III)	Rapid spasm in skeletal muscle
(D)	Myasthenia gravis	(IV)	Bone mass decreased

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-I, C-II, D-IV
- (4) A-II, B-IV, C-I, D-III

141. Choose the **correct** option for given diagram:



- (1) B $\Rightarrow$ Z-Line – Provide attachment to actin filament
- (2) A $\Rightarrow$ A-Band – Reduced during contraction
- (3) C $\Rightarrow$ Troponin – Pulls the actin filament/thin filament
- (4) D $\Rightarrow$ Myosin – Thin filament

142. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Smooth muscles does not contains alternate light and dark band.

Reason R: Smooth muscles are fusiform.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R true and R is the correct explanation of A.
- (4) Both A and R true and R is not the correct explanation of A.

143. Read the following statements (A-E):

- A. Duramater is thick, strong and fibrous layer.
- B. Piamater is a transparent membrane.
- C. Cranial meninges of brain is an epithelial tissue.
- D. Arachnoid mater is a key component of the mammalian meninges.
- E. Cerebrospinal fluid provides buoyancy to the all parts of the body.

How many of the above statements are **correct**?

- (1) Four
- (2) Three
- (3) Two
- (4) Five

144. Which of the following functions are mainly controlled by the limbic system?

- A. Emotional reaction
- B. Sexual behaviour
- C. Respiration
- D. Olfaction
- E. Body balance

Choose the **correct** option.

- (1) A, B, C, D and E
- (2) A, B and D only
- (3) A, B, C and D only
- (4) A, B, D and E only

145. Which of the following characteristic **correctly** describes the relative positions and articulations of the bones in the human hindlimb?
- (1) Fibula articulates directly with the femur and lies medial to the tibia.
 - (2) Tarsals form the digits, while phalanges form the ankle.
 - (3) Tibia articulates with the femur to form the knee joint and lies medial to the fibula.
 - (4) Femur articulates distally with the fibula at the ankle joint.
146. Which of the following characteristics are associated with locomotion?
- A. All locomotion are movements.
 - B. All movements are not locomotions.
 - C. Methods of locomotion vary with animals habitats.
 - D. Locomotion is used to escape from predators.
- (1) A and B only
 - (2) B, C and D only
 - (3) A, B and C only
 - (4) A, B, C and D
147. Which of the following cranial bone pairings is **incorrectly** matched with respect to its number in the human skull?
- (1) Parietal bone - 2 in number
 - (2) Ethmoid bone - 2 in number
 - (3) Occipital bone - 1 in number
 - (4) Frontal bone - 1 in number
148. Select the **mismatched** option.
- (1) Cartilaginous joints - Permits limited movements
 - (2) Glenoid cavity - A depression in scapula
 - (3) LMM - Has binding sites for ATP
 - (4) Clavicle - Collar bone
149. Which of the following statements is **incorrect** regarding the differences between ciliary and muscular movement in humans?
- (1) Ciliary movement is performed by specialized cells in the lining of the trachea and fallopian tubes.
 - (2) Muscular movement is essential for locomotion and changes in body posture.
 - (3) Both movements are energy-dependent and utilize ATP.
 - (4) Ciliary movement involves the contraction of sarcomeres, while muscular movement involves microtubules.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The potential difference across the membrane of neuron at rest is called resting membrane potential.
Reason R: The resting potential in a neuron is sustained solely through active transport mechanisms.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both A and R are true and R is the correct explanation of A.
 - (2) Both A and R are true but R is NOT the correct explanation of A.
 - (3) A is true but R is false.
 - (4) A is false but R is true.
151. If a chemical agent specifically blocks the Ca^{2+} -ATPase pumps on the sarcoplasmic reticulum membrane, what would be the initial effect on the skeletal muscle fibre after a single stimulation?
- (1) The muscle would not be able to contract.
 - (2) The muscle would contract but relaxation would be significantly delayed.
 - (3) The myosin heads would be unable to bind to actin.
 - (4) The action potential would not propagate down the T-tubules.
152. Cardiac muscle tissue is highly resistant to fatigue. Which of the following cellular feature is **most** directly responsible for this property?
- (1) The presence of intercalated discs with gap junctions.
 - (2) Its involuntary nature controlled by the autonomic nervous system.
 - (3) A large number of mitochondria and a high density of capillaries.
 - (4) The branching pattern of the muscle cells.

153. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Sympathetic nerve	(I)	Promotes non emergency function
(B)	Para sympathetic nerve	(II)	Mobilises body system during emergency situation
(C)	Axon	(III)	Nerve impulse travel in centrifugal manner
(D)	Dendrites	(IV)	Nerve impulse travel in centripetal manner

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-I, B-II, C-IV, D-III

154. Read the following statements about joints in the human skeletal system:

- A. The joint between the atlas and occipital condyles of the skull allows the yes movement of the head.
- B. The joint between the atlas and axis vertebrae is a pivot joint, responsible for the no movement of the head.
- C. The elbow joint is a synovial hinge joint permitting uniaxial movement.
- D. The pubic symphysis is a synovial joint with a wide range of motion.
- E. Sutures of the skull are fibrous joints and are immovable.

How many of the above statements is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

155. Given below are two statements:

Statement I: Cerebellum is the area of processing and analysis of various sensation such as touch and pain.

Statement II: Cerebrum maintain the human behaviour, creative ideas and ability of decision making.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

156. Consider the following statements related to the bones and features of the human skull:

- A. The frontal bone forms the forehead, contributes to the roof of the orbit, and protects the frontal lobe of the brain.
- B. The maxilla forms the upper jaw, contains alveolar sockets for upper teeth, and contributes to the floor of the orbit.
- C. The zygomatic bone forms the prominence of the cheek and contributes to the lateral wall and floor of the orbit.
- D. The temporal bone houses the external auditory canal, articulates with the mandible, and encloses the middle and inner ear.
- E. The occipital bone forms the anterior part of the skull, articulating with the nasal bones and housing the pituitary fossa.

Which of the above statements are anatomically **correct**?

- (1) A, B, and E only
- (2) A, C, and D only
- (3) B, C, and E only
- (4) A, B, C, and D only

157. Read the following statements:

- A. Nervous system provides an organised network of point-to-point connection.
- B. The endocrine system provides chemical integration through hormone.
- C. The neural organisation is very complex in lower invertebrates.
- D. Neuron can detect & receive stimuli but cannot transmit.

How many of the above statements is/are **false**?

- (1) Four
- (2) Three
- (3) Two
- (4) One

158. Which of the following statements is **correct** regarding human muscle tissue?

- (1) All muscle tissue in the body is derived from the ectoderm.
- (2) Muscle tissue constitutes about 10-20% of the total body weight in a healthy adult.
- (3) Skeletal muscle is a major contributor to body heat production.
- (4) All muscle contractions in the body are under voluntary control.

159. During which of the following electrical events in a neuron does the inner surface of the plasma membrane temporarily become positively charged relative to the outside?

- (1) Polarisation
- (2) Depolarisation
- (3) Repolarisation
- (4) Hyperpolarisation

160. Which of the following statements about the components of a sarcomere is **incorrect**?

- (1) The thin filament is composed of two helically wound F-actin strands.
- (2) The thick filament is composed of myosin molecules arranged in a polymeric form.
- (3) Troponin is a complex of three proteins that regulates the binding of actin and myosin.
- (4) Tropomyosin is a calcium-binding protein that initiates contraction.

161. Match List-I with List-II.

List-I (Movement)		List-II (Example)	
(A)	Amoeboid movement	(I)	Phagocytic action of a neutrophil
(B)	Ciliary movement	(II)	Contraction of the biceps muscle
(C)	Muscular movement	(III)	Movement within the cell of a plant like <i>Hydrilla</i>
(D)	Protoplasmic Streaming	(IV)	Passage of ova through the female reproductive tract

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
- (2) A-I, B-IV, C-III, D-II
- (3) A-IV, B-I, C-II, D-III
- (4) A-IV, B-II, C-I, D-III

162. Which of the following statements is **wrong** about the function of brain?

- (1) Hypothalamus mainly controls autonomic neural system.
- (2) Voluntary muscle activity is solely started by cerebellum.
- (3) Medulla oblongata mainly regulates involuntary activity of our body.
- (4) Thalamus acts as major coordinating centre for sensory and motor signalling.

163. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Bone is a specialized connective tissue.

Reason R: Bone has a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

164. Arrange the following components in **correct** functional order for a motor command originating in the brain to reach a gland or visceral organ:

- A. Autonomic nervous system
- B. Peripheral nervous system
- C. Central nervous system
- D. Parasympathetic division

- (1) C → B → A → D
- (2) B → C → A → D
- (3) C → A → B → D
- (4) C → D → A → B

165. Given below are two statements:

Statement I: Atlas is the first thoracic vertebra that supports the ribs.

Statement II: Acetabulum is the depression in femur where head of humerus fits.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

166. Which of the following part of the brain coordinates and maintains the rhythmic activity as the body's master clock?

- (1) Cerebral cortex
- (2) Hypothalamus
- (3) Thalamus
- (4) Midbrain

167. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Femur primarily supports the weight of the upper body while sitting.

Reason R: The femur is the strongest bone in the human body.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is NOT the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

168. Which of the following statements is **false** regarding synapse?

- (1) Synapse is formed by 2 membranes: first presynaptic membrane of synaptic knob & second postsynaptic membrane of dendrite.
- (2) Synaptic membranes always be separated by a gap called synaptic cleft.
- (3) Electrical synapse is very similar to impulse conduction along a single axon.
- (4) In chemical synapse, neurotransmitter is released and either excitatory or inhibitory potential is generated on postsynaptic membrane.

169. Where A stands for axon, D for dendrite, S for synapse, and CB for cell body, a typical sequence of structures between a receptor and an efferent organ is?

- (1) D → CB → A → S → D → CB → A
- (2) A → D → CB → S → A → D → CB
- (3) D → CB → A → S → A → CB → D
- (4) D → A → S → CB → D → A → CB

170. If all the phalanges, carpals, and metacarpals of both hands are counted together with all the phalanges, tarsals, and metatarsals of both feet, what will be the total number of bones involved?

- (1) 100
- (2) 106
- (3) 112
- (4) 120

171. Given below are two statements:

Statement I: Ion channels are proteins that allow passive movement of ions down their electrochemical gradients in nerve cell.

Statement II: One of the fundamental properties of the nervous system is its ability to detect and respond to stimuli (sensitivity).

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

172. Match List-I with List-II.

List-I (Bone)		List-II (Type)	
(A)	Sternum	(I)	Irregular Bone
(B)	Humerus	(II)	Short Bone
(C)	Vertebra	(III)	Flat Bone
(D)	Carpal	(IV)	Long Bone

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-I, D-IV

173. Choose the option showing **incorrect** statements regarding vertebral column.

- A. Our vertebral column is formed by 36 serially arranged units called vertebrae.
- B. It is ventrally placed.
- C. It extends from the base of the skull and constitutes the main framework of the trunk.
- D. First vertebra is the axis and it articulates with the occipital condyles.

- (1) A, B, C and D
- (2) A, B and D only
- (3) C only
- (4) B and C only

174. A 60-year-old woman is diagnosed with severe osteoporosis. Her physician explains that declining estrogen levels after menopause are a major cause. The mechanism involves estrogen's role in regulating bone remodeling. Specifically, estrogen deficiency leads to:

- (1) increased activity of osteoblasts and decreased activity of osteoclasts.
- (2) decreased absorption of calcium from the diet.
- (3) increased activity of bone resorption and decreased activity of bone formation.
- (4) the conversion of bone marrow to fatty tissue.

175. Match List-I with List-II:

List-I		List-II	
(A)	Corpora quadrigemina	(I)	The inner parts of cerebral hemispheres, amygdala and hippocampus
(B)	Spinal cord	(II)	Four round swellings in the midbrain
(C)	Cerebral aqueduct	(III)	Attached to medulla
(D)	Limbic system	(IV)	Canal passing through the midbrain

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-I, C-II, D-III

176. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The 11th and 12th pairs of ribs are called floating ribs.

Reason R: Floating ribs allow greater movement of only the diaphragm during deep breathing.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

177. How many of the following statements is/are **correct** about synovial joints?

- A. They contain a fluid-filled cavity.
- B. They allow minimal movement only.
- C. Found in thumb and knee.
- D. They include ball-and-socket joints.

- (1) Only one
- (2) Only two
- (3) Only three
- (4) All four

178. Pick up the odd one in each series and select **correct** option.

- A. Actin, Myosin, Sarcomere, Troponin.
- B. Fascicles, tropomyosin, Muscle fibre, Myofibril.
- C. Myosin, Actin, Troponin, Tropomyosin.
- D. ATPase enzyme, ATP binding site, Actin binding site, Myosin binding site.

	A	B	C	D
(1)	Sarcomere	Tropomyosin	Myosin	Myosin binding site
(2)	Sarcomere	Tropomyosin	Myosin	Actin binding site
(3)	Troponin	Fascicles	Actin	ATP binding site
(4)	Troponin	Fascicles	Tropomyosin	ATPase enzyme

179. Which of the following **best** describes the function of the association areas of cortex?

- (1) They generate motor impulses for voluntary skeletal movements.
- (2) They receive direct sensory input from only receptors in sense organs.
- (3) They integrate information from different sensory modalities and are involved in reasoning, learning, and communication.
- (4) They control reflex actions and maintain balance through the cerebellum.

■■■

180. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Voluntary muscle fibre	(I)	Branched, uninucleated, with intercalated discs
(B)	Cardiac muscle fibre	(II)	Cylindrical, multinucleated, striated
(C)	Visceral muscle fibre	(III)	Bundles of skeletal muscle fibres wrapped in perimysium
(D)	Fascicle	(IV)	Spindle-shaped, uninucleated, involuntary

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-III, B-II, C-I, D-IV



SPACE FOR ROUGH WORK

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____